

ShopFlow AWS Project

SSM Parameter Store Design

Config Strings - Feature Flags - Cache TTLs - Rate Limits

Module 5 of 18

Field	Value
Version	1.0
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Region	ap-south-1 (Mumbai)
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Application	ShopVerse -- Spring Boot 3.2.5 / Java 21

1. Overview

AWS Systems Manager Parameter Store is a centralised store for application configuration strings. ShopFlow uses it for all non-sensitive config: database hostnames, Redis endpoints, cache TTL values, S3 bucket names, feature flags, and rate limit thresholds. Parameters are organised under a hierarchical path prefix: `/shopflow/{environment}/{service}/{key}`. All Standard parameters are free.

2. Database Parameters

Parameter Path	Value	Updated When
/shopflow/dev/db/endpoint	FILL_AFTER_RDS_DEPLOY	After Module 6 (RDS) deploys
/shopflow/dev/db/port	3306	Static -- never changes
/shopflow/dev/db/name	shopverse	Static -- matches Flyway schema

3. Redis Parameters

Parameter Path	Value	Notes
/shopflow/dev/redis/endpoint	FILL_AFTER_ELASTICACHE_DEPLOY	After Module 7 deploys
/shopflow/dev/redis/port	6379	Default Redis port
/shopflow/dev/redis/ttl/products	1800	30 min -- matches CacheConfig.java
/shopflow/dev/redis/ttl/customers	3600	60 min
/shopflow/dev/redis/ttl/orders	300	5 min -- orders change frequently

4. Application Parameters

Parameter Path	Value	Notes
/shopflow/dev/app/port	8080	Spring Boot server port
/shopflow/dev/app/environment	dev	Maps to spring.profiles.active
/shopflow/dev/app/log-level	INFO	Change to DEBUG for troubleshooting
/shopflow/dev/app/jwt-expiry-seconds	86400	24-hour token lifetime

5. Feature Flags

Parameter Path	Default	Effect When false
/shopflow/dev/feature/graphql-enabled	true	Disables ProductGraphQLController endpoint
/shopflow/dev/feature/reactive-enabled	true	Disables ReactiveProductController

Parameter Path	Default	Effect When false
/shopflow/dev/feature/maintenance-mode	false	When true: all writes return 503

6. Rate Limiting

Parameter Path	Value	Notes
/shopflow/dev/rate-limit/auth-per-minute	10	Max login attempts per IP
/shopflow/dev/rate-limit/api-per-minute	200	Max API requests per authenticated user

7. Quick Reference -- Update After Each Module

After Module	Update Command
6 -- RDS	aws ssm put-parameter --name /shopflow/dev/db/endpoint --value <endpoint> --overwrite
7 -- ElastiCache	aws ssm put-parameter --name /shopflow/dev/redis/endpoint --value <endpoint> --overwrite